

Connecting to Azure Cosmos DB Table API using the .NET SDK

Summary:-

In this lab, you will create a .NET console application to connect to an Azure Cosmos DB Table API account. You will install the Azure Tables SDK, configure the application using the account endpoint and access key, and prepare the project for performing Table API operations.

Let's Start:-

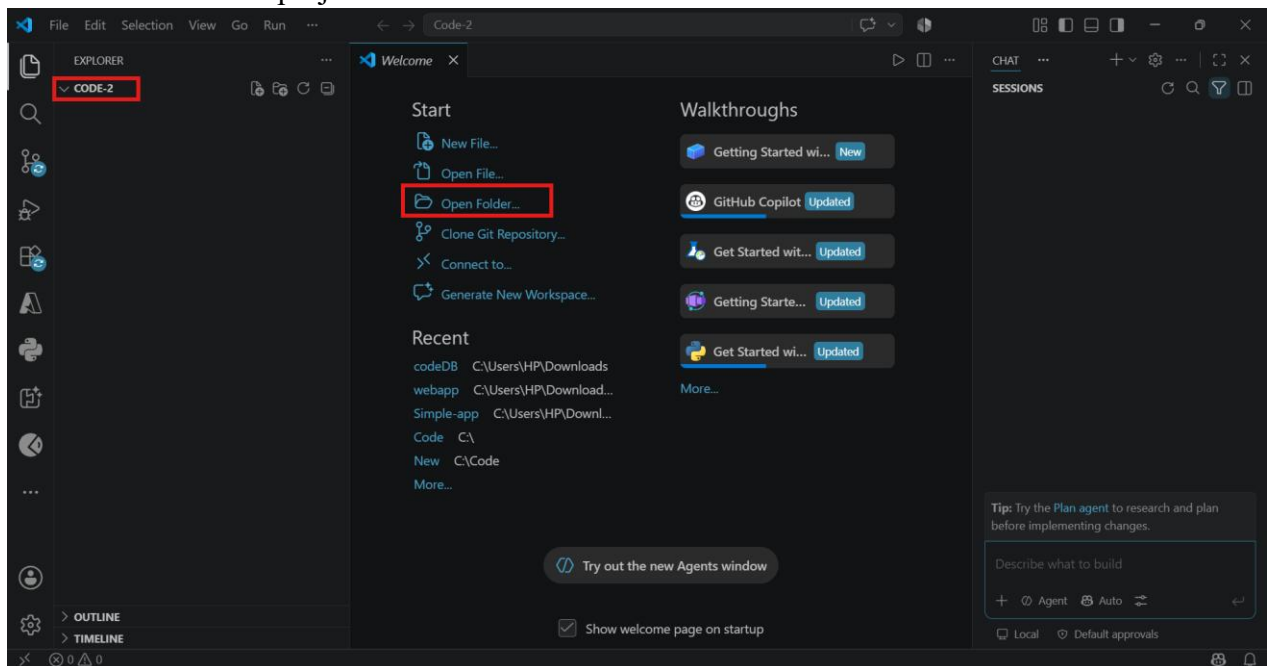
After completing this lab, you will be able to:

- Create a .NET console application.
- Install the Azure Tables SDK.
- Configure the application to connect to an Azure Cosmos DB Table API account.
- Build and run the application successfully.

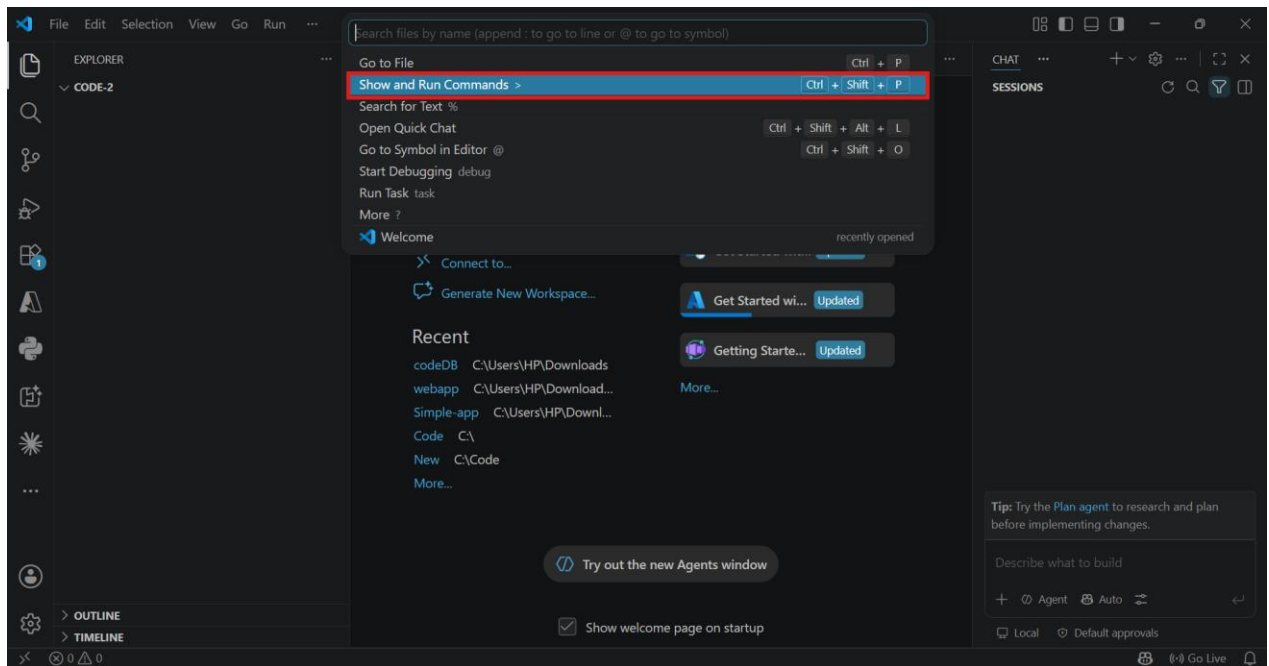
Lab Steps:-

Step 1 – Create a .NET Console Application

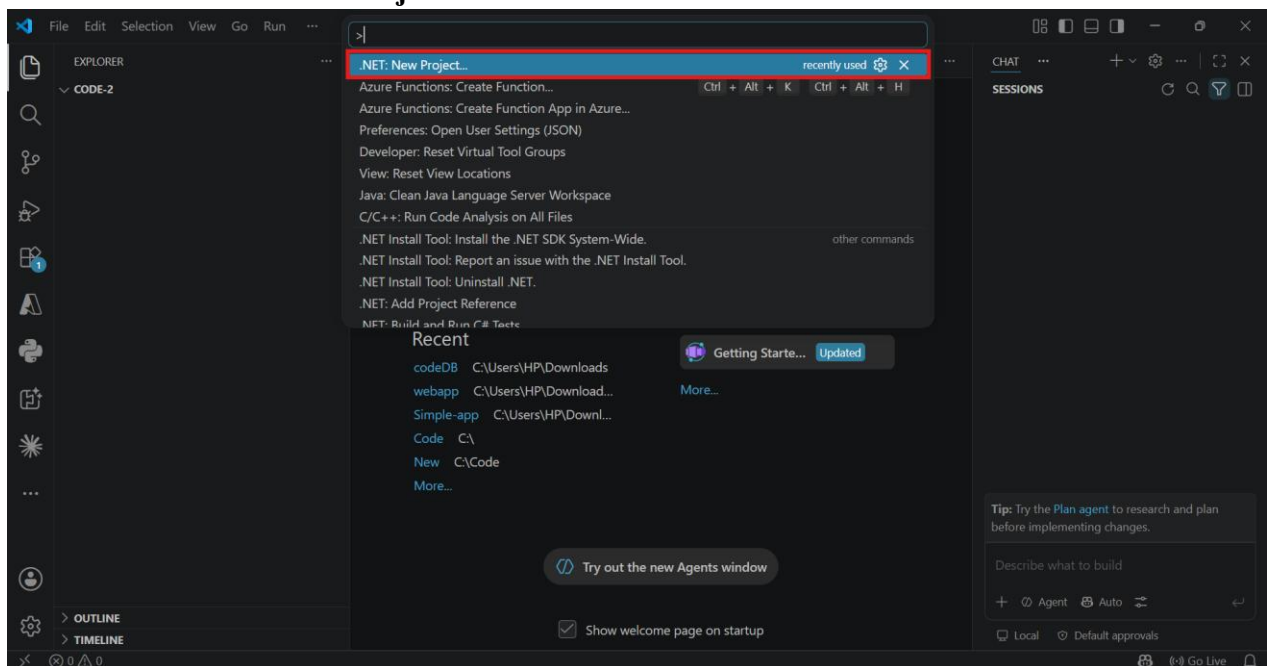
- Open **Visual Studio Code**.
- Create a new project folder.



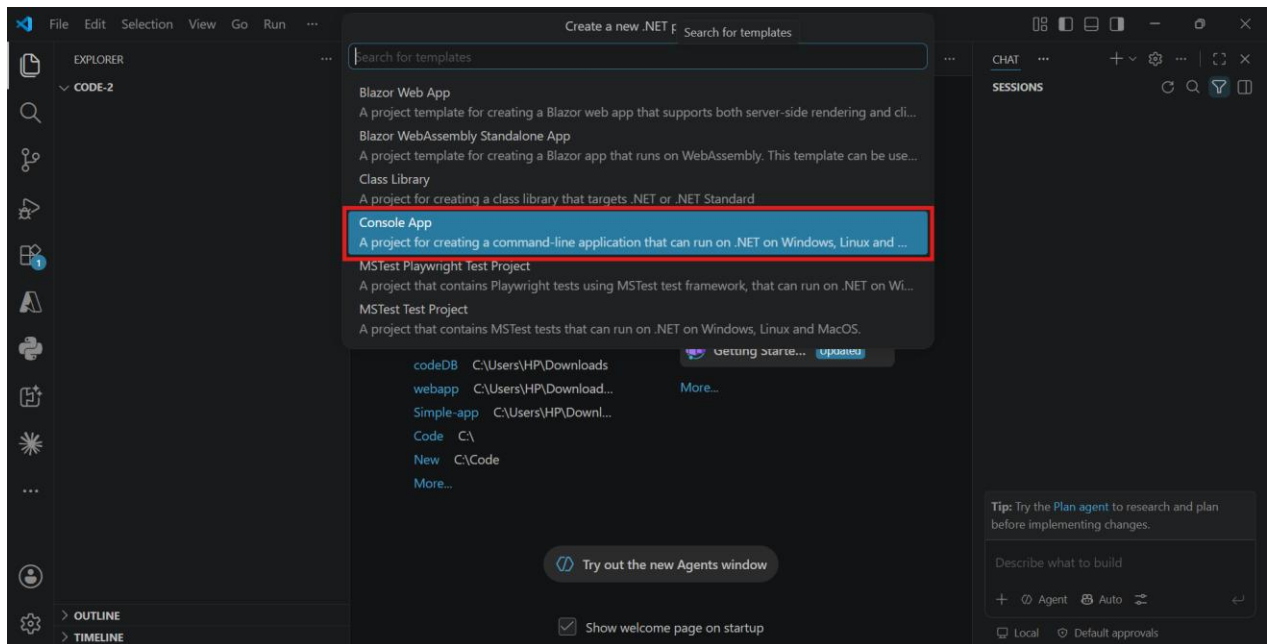
- Press **Ctrl + Shift + P** to open the Command Palette.
- Select **Show and Run Commands**.



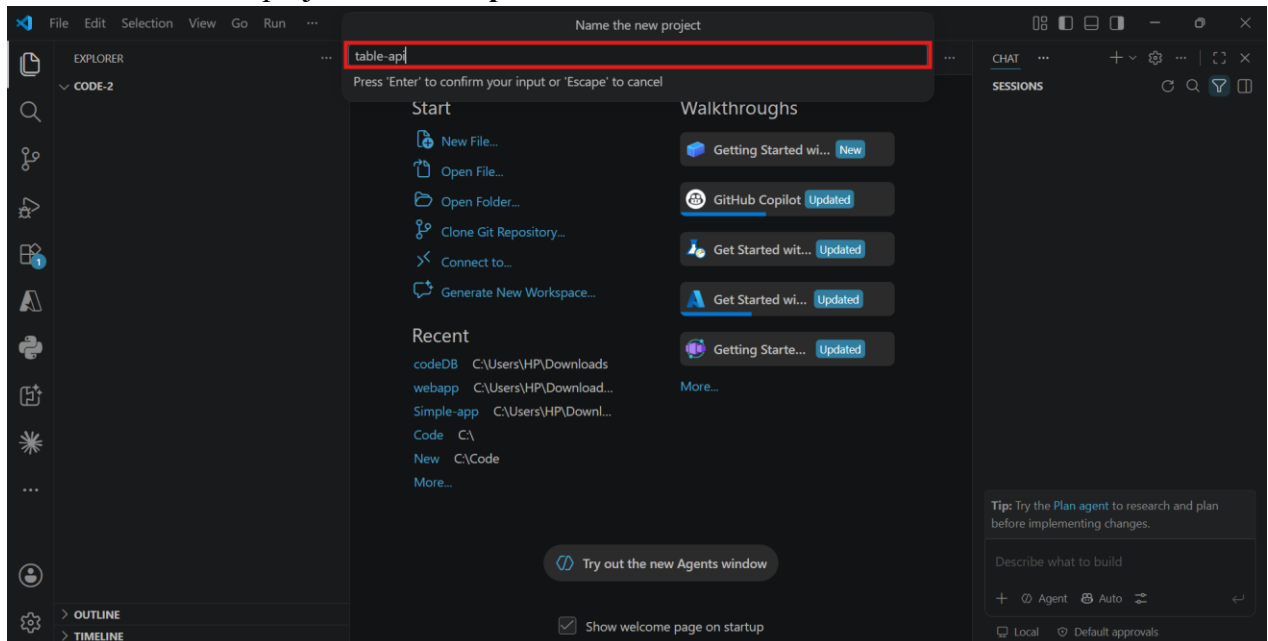
- **Select .NET: New Project.**



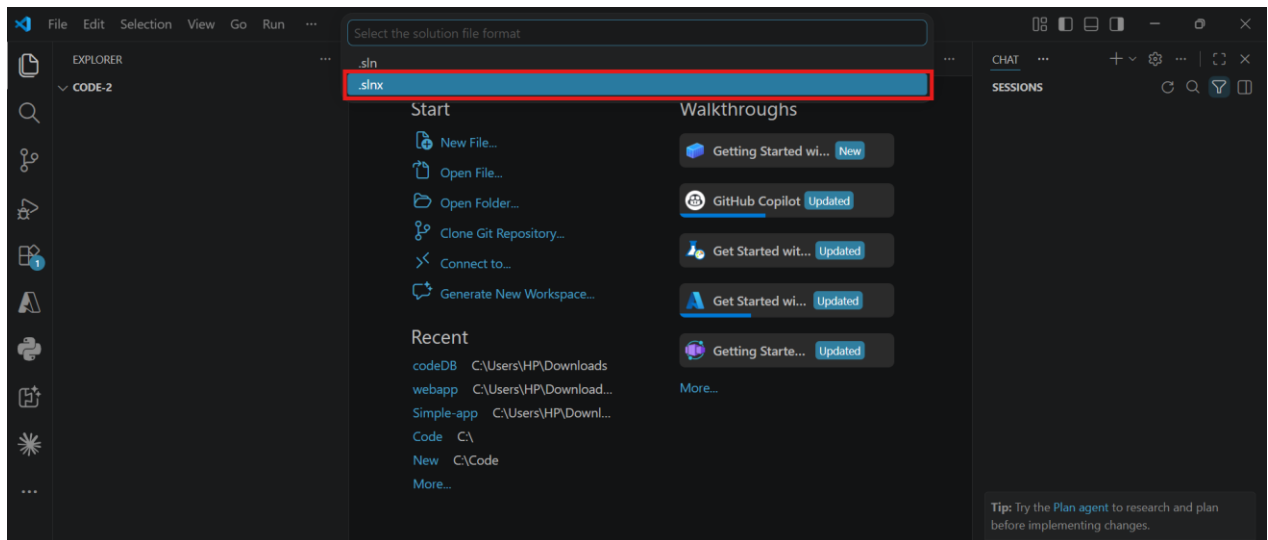
- **Select Console App.**



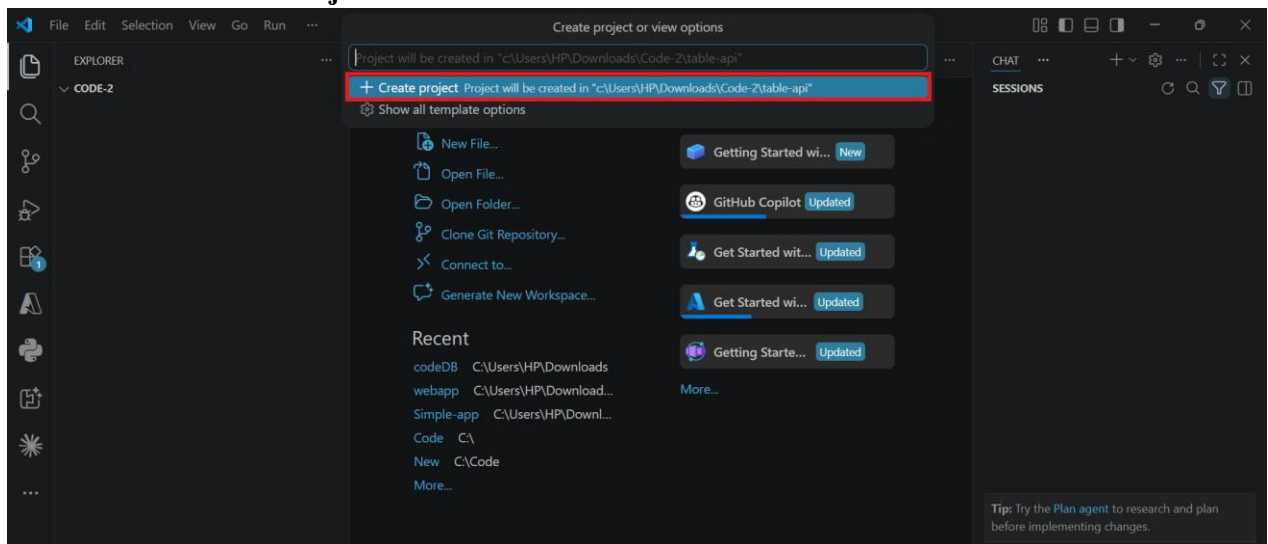
- Rename the project as **table-api**.



- Select the default directory. Create the **.slnx** solution file.



- Click **Create Project**.



- This step creates a new .NET console application.

Step 2 – Install the Azure Tables SDK

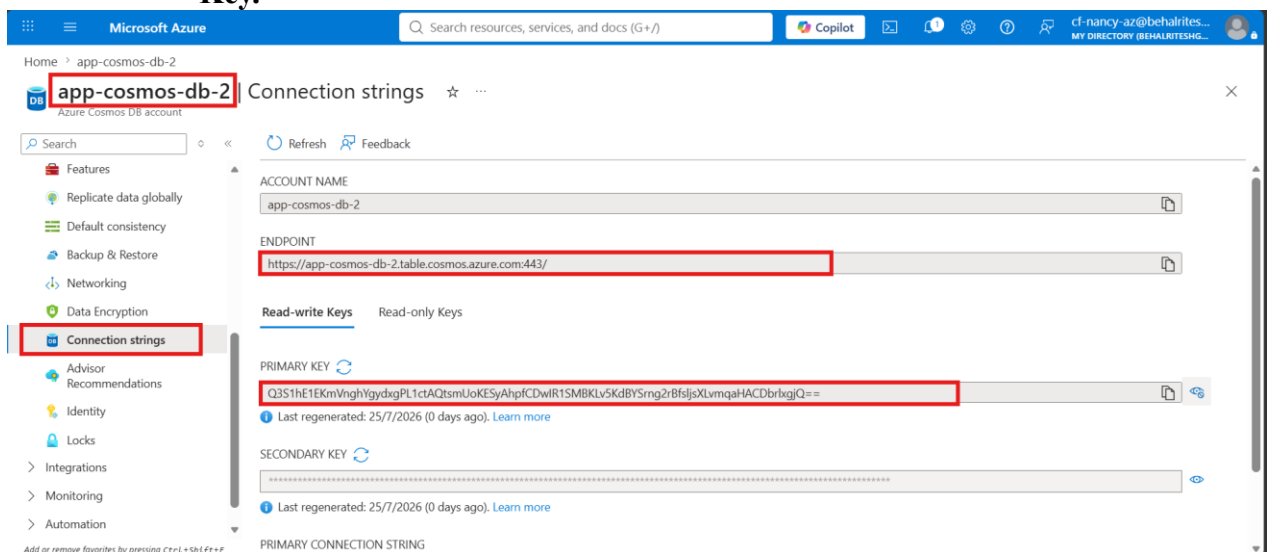
- **Open** the integrated terminal.
- **Run** the following commands:
cd .\tableapi-app\
dotnet add package Azure.Data.Tables

The screenshot shows the Visual Studio Code interface. The Explorer pane on the left shows a project named 'table-api' with files 'table-api.csproj' and 'table-api.slnx'. The main editor shows 'Program.cs' with the code: `1 Console.WriteLine("Hello, World!");`. The bottom terminal window shows the command `cd .\table-api\` and `dotnet add package Azure.Data.Tables` being executed. The output shows the package being restored and installed successfully.

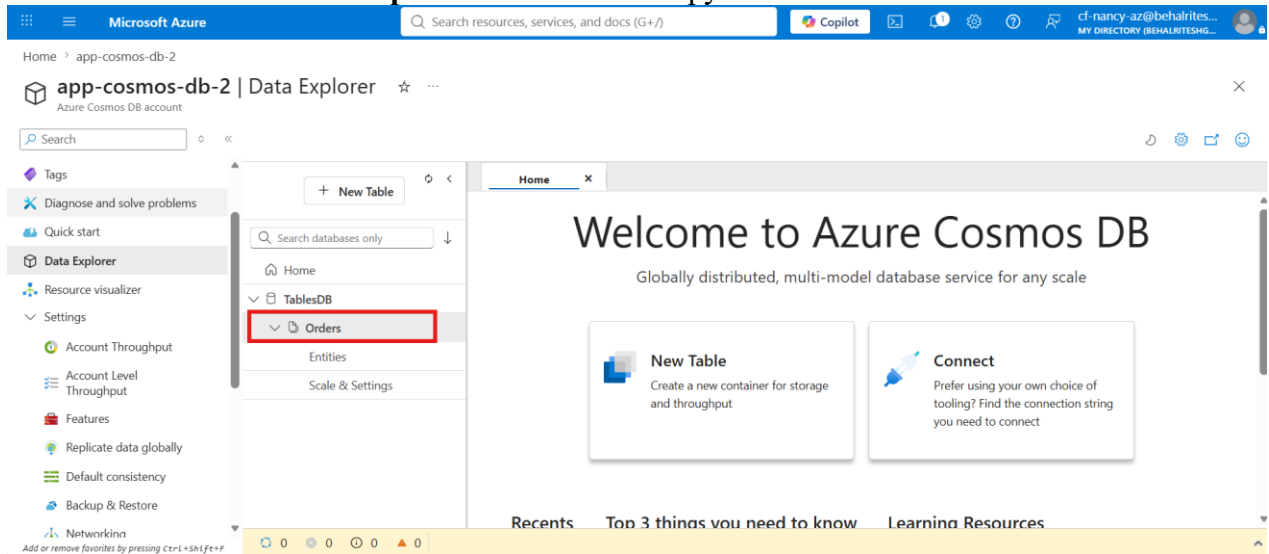
- Wait for the **package installation** to complete successfully.
- This step installs the Azure Tables .NET SDK required to interact with Azure Cosmos DB Table API.

Step 3 – Get the Connection Details

- Sign in to the **Azure Portal**.
- Open the **Azure Cosmos DB Table API** account.
- Copy the following values:
 - **Endpoint**
 - **Account Name**
 - From the left pane, select **Connection Strings**. Copy **Primary Key/Secondary Key**.



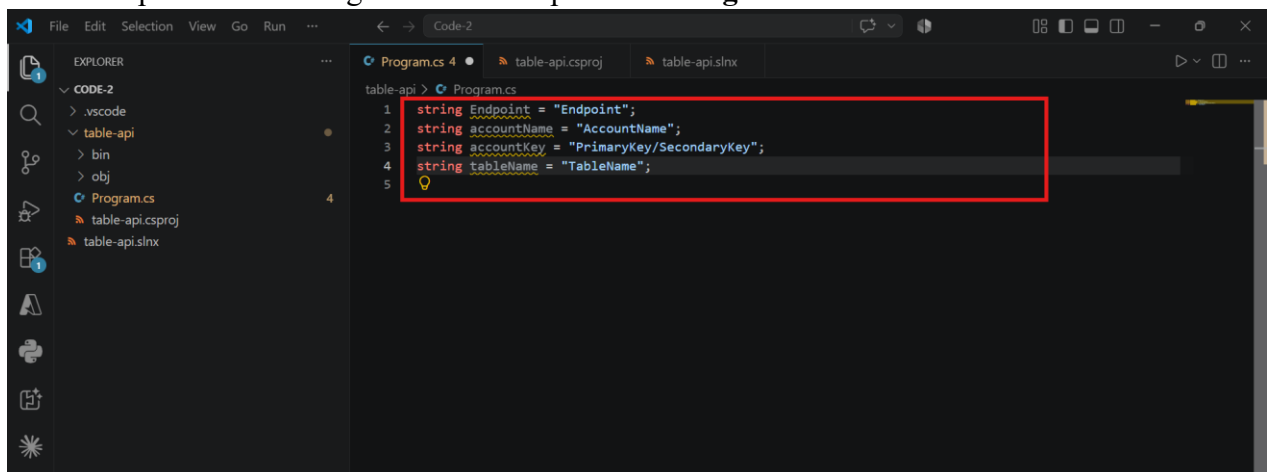
- Go to **Data Explorer > Database**. Copy Table Name.



- This step retrieves the connection details required for the application.

Step 4 – Add the Project Files

- Open **Program.cs**.
- Replace the existing code with the provided **Program.cs** file.



- Paste** the copied values in the code.

```
1
2
3 string Endpoint = "https://app-cosmos-db-2.table.cosmos.azure.com:443/";
4 string accountName = "app-cosmos-db-2";
5 string accountKey = "Q3S1hE1EKmVnghYgydxgPL1ctAQtsmUoKESyAhpFCdWIR1SMBKLv5KdBYsRng2rBfsljsXLvmqaHACDb1vg3Q==";
6 string tableName = "Orders";
7
8
9
10
11
12
13
14
15
16
17
18
19
20
```

- Add the provided **Program.cs** code.

```
1 using System;
2 using System.Threading.Tasks;
3 using Azure;
4 using Azure.Data.Tables;
5
6 string Endpoint = "https://app-cosmos-db-2.documents.azure.com:443/";
7 string accountName = "app-cosmos-db-2";
8 string accountKey = "Q3S1hE1EKmVnghYgydxgPL1ctAQtsmUoKESyAhpFCdWIR1SMBKLv5KdBYsRng2rBfsljsXLvmqaHACDb1vg3Q==";
9 string tableName = "Orders";
10
11
12 var serviceClient = new
13 TableServiceClient(new Uri(Endpoint), new TableSharedKeyCredential(accountName, accountKey));
14
15 TableClient tableClient = serviceClient.GetTableClient(tableName);
16
17 var orders = new[]
18 {
19     new TableEntity(partitionKey: "C004", rowKey: "ord-3001")
20     {
21         ["CustomerName"] = "Sara",
22         ["OrderDateUtc"] = DateTime.Parse("2026-02-11T15:30:00Z").ToUniversalTime(),
23         ["Status"] = "Paid",
24         ["CourseId"] = "DP-600",
25         ["CourseName"] = "DP-600 Fabric Analytics Engineer",
26         ["Category"] = "Data",
27         ["Method"] = "Card",
28         ["Provider"] = "Visa",
29         ["TransactionId"] = "tx-9001",
30         ["PaidAmount"] = 59.99,
31     }
32 }
```

- **Save** all the files.
- This step prepares the application for Azure Cosmos DB Table API operations.

Step 5 – Build and Run the Application

- **Open** the integrated terminal and **run**:
dotnet build
dotnet run

The screenshot shows the Visual Studio Code editor with a C# project named 'table-api'. The code in Program.cs configures the Azure Cosmos DB Table API client and inserts three orders. The terminal shows the successful build and execution of the application.

```

1 using System;
2 using Azure;
3 using Azure.Data.Tables;
4
5
6 string Endpoint = "https://app-cosmos-db-2.table.cosmos.azure.com:443/";
7 string accountName = "app-cosmos-db-2";
8 string accountKey = "Q3S1hE1EKmVnghiYgdxgPL1ctAQtsmUoKESyAhpFCdwIR1SMBKLv5KdBYsRng2rBfsljsXLvmqahACDb-r1xgJQ==";
9 string tableName = "Orders";
10
11 var serviceClient = new
12 TableServiceClient(new Uri(Endpoint), new TableSharedKeyCredential(accountName, accountKey));
13
14 TableClient tableClient = serviceClient.GetTableClient(tableName);
15
16 var orders = new[]
17 {
18     new TableEntity(partitionKey: "C004", rowKey: "ord-3001")
19 }

```

Terminal Output:

```

PS C:\Users\HP\Downloads\Code-2\table-api> dotnet build
Restore complete (1.0s)
table-api net10.0 succeeded (0.6s) -> bin\Debug\net10.0\table-api.dll
Build succeeded in 3.2s
PS C:\Users\HP\Downloads\Code-2\table-api> dotnet run
Inserted: PK=C004, RK=ord-3001
Inserted: PK=C002, RK=ord-3002
Inserted: PK=C004, RK=ord-3003
PS C:\Users\HP\Downloads\Code-2\table-api>

```

- Verify that the application executes successfully without any errors.

The screenshot shows the Azure Cosmos DB Data Explorer interface for the 'app-cosmos-db-2' account. The 'Orders' table is selected, and the data is displayed in a table format. The table contains three rows of data, which are highlighted with a red box.

PartitionKey	RowKey	Timestamp	Category	CourseId	CourseName	Currency	CustomerName	FailureReason	Method	OrderDate
C002	ord-3002	Sat, 25 Jul 2026 10:31:00 GMT	Azure	AZ-204	AZ-204 Azure Developer	USD	James	InsufficientFunds	Card	Wed, 11 F
C004	ord-3001	Sat, 25 Jul 2026 10:31:00 GMT	Data	DP-600	DP-600 Fabric Analytics Engineer	USD	Sara		Card	Wed, 11 F
C004	ord-3003	Sat, 25 Jul 2026 10:31:01 GMT	Azure	AZ-204	AZ-204 Azure Developer	USD	Sara		Card	Sat, 25 Ju

- This step builds and runs the .NET application successfully.

Verification

Verify the following:

- The .NET console application is created successfully.
- The **Azure.Data.Tables** package is installed successfully.
- The application is configured with the Azure Cosmos DB Table API connection details.
- The application connects successfully to the Azure Cosmos DB Table API account.
- The project builds and runs successfully without errors.